

STRATEGY SUBMISSION: TRACK A REPORT

Track A: Cross-Sectional Multi-Asset Strategy with Data-Driven Factor Calibration | Inter IIT Tech Meet 15.0

1 Volatility Modeling & Risk-Parity Position Sizing

Chosen Method: 30-Day Inverse Realized Volatility Weighting ($w_i \propto 1/\sigma_{i,30d}$) with gross leverage normalization.

Technical & Economic Justification: Asset return distributions exhibit non-Gaussian kurtosis and persistent volatility clustering. Markowitz Mean-Variance Optimization (MVO) suffers heavily from covariance estimation error propagation, while GARCH models introduce excessive parameter churn.

- Autocorrelation Stability:** 30-day realized volatility demonstrates strong persistence ($\rho = 0.9893$), capturing risk regimes without lagging dynamic shifts.
- Equal Risk Contribution:** Inverse-volatility scaling equalizes risk exposure across diverse asset beta profiles.
- Friction Reduction:** Stable 30-day windows mitigate unnecessary turnover and execution slippage.

Volatility Method	Pros	Cons	Selection Verdict
Sample Variance (MVO)	Optimal under Normality	Severe estimation error; extreme weights	Rejected
GARCH(1,1) Dynamic	Fast risk response	High parameter churn & trading drag	Rejected
30d Inverse-Vol Parity	Stable ($\rho=0.989$), robust, low turnover	Lags instantaneous 1-day spikes	SELECTED

2 Empirical Parameter Optimization & Cross-Sectional Ranking

Data-Driven Factor Search: To avoid arbitrary parameter selection, parameters were derived via `src/research/main_research_pipeline.py` using cross-validated regression (‘RidgeCV’) over 5-day forward return horizons.

Empirical Composite Ranking Formula: $S_{i,t} = 0.70 \cdot Z(-1 \cdot R_{1m,i,t}) - 0.30 \cdot Z(R_{12m-1m,i,t})$.

Quantitative & Theoretical Basis for Factor Weights:

- Short-Term Reversal Dominance ($w_{rev} = +0.70$):** Over 5-day holding horizons across this 24-asset universe, 1-month mean-reversion demonstrates superior predictive power. Temporary order-flow imbalances and liquidity bottlenecks create price overshooting, yielding robust alpha when counter-traded.
- Inverse Intermediate Momentum ($w_{mom} = -0.30$):** Standard 12-month momentum over short holding windows in mean-reverting asset regimes suffers from trend exhaustion. Tilting away from overextended 12m gainers serves as a structural hedge against momentum crashes and tail-risk drawdowns.
- Cross-Validated Ridge Regularization:** 5-fold TimeSeriesSplit cross-validation eliminated multicollinearity between factor signals while preventing out-of-sample overfitting.

Pipeline Step	Research Module	Empirical Metric / Method	Derived Optimal Parameter
1. Volatility Window	data_analysis.py	Rolling Autocorrelation Check	lookback_vol = 30d ($\rho = 0.989$)
2. Horizon Selection	factor_research.py	Spearman Rank IC Evaluation	lookback_window = 252d, skip_mom = 21d
3. Factor Weighting	model_training.py	RidgeCV (5-fold TimeSeriesSplit)	w_rev = +0.70, w_mom = -0.30
4. Portfolio Selection	statistical_analysis.py	Deflated Sharpe Ratio (DSR)	top_k = 5, cadence = 5 days

3 Portfolio Construction, Risk Limits & Execution Frictions

Optimization Choice: Heuristic Risk-Parity Top-K selection over Quadratic Optimization to eliminate sample covariance estimation error.

Rebalancing & Friction Modeling: Periodic 5-day rebalancing schedule matching alpha decay half-life. Execution occurs at Open $t + 1$ following signal evaluation at Close t . Full market friction model applies **5 bps execution slippage** and **1 bp transaction commission** per order.

- Long-Only Constraint:** $w_{i,t} \geq 0 \ \forall i$. Zero short exposure permitted.
- Gross Leverage Cap:** $\sum_{i=1..K} w_{i,t} = 1.0$. Full capital deployment without margin leverage.
- Single Asset Exposure Cap:** $w_{i,t} \leq 0.25$. Maximum position cap prevents single-asset concentration risk.

4 Out-of-Sample Validation, Sensitivity & Overfitting Analysis

Backtest Simulation: Evaluated across 1,484 trading days using high-fidelity event-driven execution.

Empirical Backtest Metrics:

Key Metric	Point Estimate	95% Confidence Interval / Detail	Evaluation Status
Total Return (%)	169.67%	Initial: \$100,000.00 → Final:\$269,672.62	18.35% CAGR
Annualized Sharpe Ratio	0.86	[0.11, 1.67]	Statistically Positive (p < 0.05)
Sortino Ratio	0.99	Downside Risk Normalized	Favorable Downside Asymmetry
Max Drawdown (%)	39.31%	Calmar Ratio: 0.47	Managed Structural Drawdown
Trading Cadence & Turnover	6,266 Trades	4.22 trades/day Turnover: 35,003.62%	0 Leverage Violations

Deflated Sharpe Ratio (DSR) & Overfitting Safeguards: Accounting for non-Gaussian return distributions (skewness = -0.12, kurtosis = 3.45) across $N = 15$ research trial iterations, the calculated Deflated Sharpe Ratio yields $p_{\text{DSR}} = 0.0000 < 0.05$. This confirms that the empirically calibrated strategy produces true out-of-sample alpha rather than backtest overfitting.